

Airpay Academy 2.0 — Complete Installation Guide

From Bare Server to Running LMS

Version: v1.0.0-rc1 | **Date:** April 2026 **Audience:** IT Operations / System Administrator **Estimated Time:** 2-4 hours for fresh install, 1 hour for upgrade

Table of Contents

- 1. Server Requirements
- 2. Fresh Installation (new server)
- 3. Upgrade from Production (existing server)
- 4. Post-Install Configuration
- 5. Verification Checklist
- 6. Rollback Procedure
- 7. Troubleshooting

1. Server Requirements

Hardware (Minimum)

Component	Minimum	Recommended
CPU	2 cores	4+ cores
RAM	4 GB	8+ GB
Disk	20 GB	50+ GB (SCORM content grows)
Network	10 Mbps	100+ Mbps

Software Stack

Component	Required Version	Production Current
Operating System	Linux (Ubuntu 22.04+) or Windows Server	AWS Linux
Web Server	Apache 2.4+ or Nginx	Apache
PHP	8.1 - 8.3	8.2.12
Database	MySQL 8.0+ or MariaDB 10.6+	MySQL 8.0.44 (AWS RDS)
SSL	Required (HTTPS)	Yes (ACM cert)

Required PHP Extensions

```
php-curl php-gd php-intl php-json php-mbstring php-mysql  
php-openssl php-soap php-xml php-xmlrpc php-zip php-sodium
```

DNS and Domain

- Domain: www.airpay.academy (already configured)
- SSL certificate: AWS Certificate Manager (already configured)
- DNS: Route 53 pointing to EC2/ELB (already configured)

2. Fresh Installation (New Server)

Use this section ONLY if setting up a completely new server. If upgrading the existing production server, skip to Section 3.

Step 1: Prepare the Server

```
# Ubuntu/Debian
sudo apt update && sudo apt upgrade -y
sudo apt install -y apache2 php8.2 php8.2-
{curl,gd,intl,json,mbstring,mysql,opcache,soap,xml,xmldr,zip,sodium} libapache2-mod-php8.2
unzip

# Enable Apache modules
sudo a2enmod rewrite ssl headers
sudo systemctl restart apache2
```

Step 2: Create Database

```
# Connect to MySQL (or AWS RDS)
mysql -h lms-prod-db.crpst4qn6rtu.ap-south-1.rds.amazonaws.com -u admin -p

# Create database
CREATE DATABASE airpayprod DEFAULT CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci;
CREATE USER 'moodleuser'@'%' IDENTIFIED BY '<STRONG_PASSWORD>';
GRANT ALL PRIVILEGES ON airpayprod.* TO 'moodleuser'@'%';
FLUSH PRIVILEGES;
```

Step 3: Extract Release Package

```
cd /var/www/html
# Extract the full release package
tar -xzf /path/to/airpay-academy-v1.0.0-rc1-full.tar.gz

# The package extracts to the current directory with all Moodle directories:
# admin/ auth/ blocks/ course/ enrol/ lib/ local/ mod/ theme/ etc.

# Set permissions
sudo chown -R www-data:www-data /var/www/html/
sudo chmod -R 755 /var/www/html/
```

Step 4: Create moodledata Directory

```
sudo mkdir -p /var/moodledata
sudo chown www-data:www-data /var/moodledata
sudo chmod 770 /var/moodledata
```

Step 5: Create config.php

```
sudo cp /var/www/html/config-dist.php /var/www/html/config.php
sudo nano /var/www/html/config.php
```

Edit config.php with these values:

```
$CFG->dbtype      = 'mysqli';
$CFG->dbhost       = 'lms-prod-db.crpst4qn6rtu.ap-south-1.rds.amazonaws.com';
$CFG->dbname       = 'airpayprod';
$CFG->dbuser       = 'moodleuser';
$CFG->dbpass       = '<DB_PASSWORD>';
$CFG->prefix       = 'mdl_';
$CFG->wwwroot      = 'https://www.airpay.academy';
$CFG->dataroot     = '/var/moodledata';
$CFG->directorypermissions = 0770;
```

Step 6: Import Database (if restoring from backup)

```
# If you have the production database backup:
mysql -h <RDS_HOST> -u moodleuser -p airpayprod < airpayprod-6-4.sql

# Update wwwroot if it was set to localhost during dev:
mysql -h <RDS_HOST> -u moodleuser -p airpayprod -e "
    UPDATE mdl_config SET value = 'https://www.airpay.academy' WHERE name = 'wwwroot';
"
```

Step 7: Run Moodle Installer

If this is a fresh database (no import):

```
sudo -u www-data php /var/www/html/admin/cli/install.php \
  --wwwroot=https://www.airpay.academy \
  --dataroot=/var/moodledata \
  --dbtype=mysqli \
  --dbhost=<RDS_HOST> \
  --dbname=airpayprod \
  --dbuser=moodleuser \
  --dbpass=<DB_PASSWORD> \
  --fullname="Airpay Academy" \
  --shortname="airpay" \
  --adminuser=admin \
```

```
--adminpass=<ADMIN_PASSWORD> \  
--adminemail=academy@airpay.co.in \  
--agree-license
```

Step 8: Run Post-Deploy SQL

```
mysql -h <RDS_HOST> -u moodleuser -p airpayprod < /var/www/html/config/post-deploy.sql
```

Step 9: Purge Caches

```
sudo -u www-data php /var/www/html/admin/cli/purge_caches.php
```

Step 10: Configure Apache VirtualHost

```
<VirtualHost *:443>  
    ServerName www.airpay.academy  
    DocumentRoot /var/www/html  
  
    SSLEngine on  
    SSLCertificateFile /path/to/cert.pem  
    SSLCertificateKeyFile /path/to/key.pem  
  
    <Directory /var/www/html>  
        AllowOverride All  
        Require all granted  
    </Directory>  
  
    ErrorLog ${APACHE_LOG_DIR}/academy-error.log  
    CustomLog ${APACHE_LOG_DIR}/academy-access.log combined  
</VirtualHost>
```

3. Upgrade from Production (Existing Server)

Use this section to upgrade the existing www.airpay.academy from epsilon theme to airpayux.

Pre-Upgrade Checklist

- ☐ Full database backup taken
- ☐ Full file backup of /var/www/html/ taken
- ☐ Full backup of /var/moodledata/ taken
- ☐ Maintenance mode enabled: Site Admin > Server > Maintenance mode
- ☐ Cron job paused
- ☐ SMTP verified working (or will configure after upgrade)

Step 1: Enable Maintenance Mode

```
sudo -u www-data php /var/www/html/admin/cli/maintenance.php --enable
```

Step 2: Backup Current Installation

```
# Database
mysqldump -h <RDS_HOST> -u moodleuser -p --single-transaction airpayprod >
/backup/airpayprod-pre-upgrade-$(date +%Y%m%d).sql

# Files
sudo tar -czf /backup/moodle-files-pre-upgrade-$(date +%Y%m%d).tar.gz /var/www/html/
sudo tar -czf /backup/moodledata-pre-upgrade-$(date +%Y%m%d).tar.gz /var/moodledata/
```

Step 3: Extract Release Package

```
cd /tmp
tar -xzf /path/to/airpay-academy-v1.0.0-rc1-full.tar.gz -C /tmp/academy-release/

# Copy theme (new)
sudo cp -r /tmp/academy-release/theme/airpayux /var/www/html/theme/

# Copy custom plugins (new)
sudo cp -r /tmp/academy-release/local/airpay_pages /var/www/html/local/
sudo cp -r /tmp/academy-release/local/airpay_lifecycle /var/www/html/local/
sudo cp -r /tmp/academy-release/local/airpay_integrations /var/www/html/local/
sudo cp -r /tmp/academy-release/blocks/airpay_compliance /var/www/html/blocks/

# Apply BizLMS fixes (overlay on existing files)
# jQuery AMD fixes for 10 mustache templates
sudo cp /tmp/academy-release/local/costcenter/templates/global_filter.mustache
/var/www/html/local/costcenter/templates/
```

```

sudo cp /tmp/academy-release/local/users/templates/skillprofiletabs.mustache
/var/www/html/local/users/templates/
sudo cp /tmp/academy-release/local/users/templates/skillinfotabs.mustache
/var/www/html/local/users/templates/
sudo cp /tmp/academy-release/local/search/templates/filters.mustache
/var/www/html/local/search/templates/
sudo cp /tmp/academy-release/local/program/templates/levelstab_content.mustache
/var/www/html/local/program/templates/
sudo cp /tmp/academy-release/local/program/templates/custom_header.mustache
/var/www/html/local/program/templates/
sudo cp /tmp/academy-release/local/groups/templates/global_filter.mustache
/var/www/html/local/groups/templates/
sudo cp /tmp/academy-release/local/classroom/templates/classroomContent.mustache
/var/www/html/local/classroom/templates/
sudo cp /tmp/academy-release/local/classroom/templates/custom_header.mustache
/var/www/html/local/classroom/templates/
sudo cp /tmp/academy-release/local/classroom/templates/classroomtabs.mustache
/var/www/html/local/classroom/templates/

# Theme nav-drawer fix
sudo cp /tmp/academy-release/theme/airpayux/templates/nav-drawer.mustache
/var/www/html/theme/airpayux/templates/

# Lang file typo fixes (9 files)
for plugin in users program onlineexams learningplan forum evaluation courses costcenter
classroom; do
    sudo cp /tmp/academy-release/local/$plugin/lang/en/local_$plugin.php
/var/www/html/local/$plugin/lang/en/
done

# Missing JS file
sudo cp /tmp/academy-release/local/courses/fullldescriptionpopover.js
/var/www/html/local/courses/

# Fixed learningplan index.php
sudo cp /tmp/academy-release/local/learningplan/index.php /var/www/html/local/learningplan/

# Fixed lifecycle messages.php
sudo cp /tmp/academy-release/local/airpay_lifecycle/db/messages.php
/var/www/html/local/airpay_lifecycle/db/

# Set permissions
sudo chown -R www-data:www-data /var/www/html/theme/airpayux
sudo chown -R www-data:www-data /var/www/html/local/airpay_*
sudo chown -R www-data:www-data /var/www/html/blocks/airpay_compliance

```

Step 4: Run Database Upgrade

```

sudo -u www-data php /var/www/html/admin/cli/upgrade.php

```

Or navigate to: <https://www.airpay.academy/admin/index.php> and click "Upgrade Moodle database now"

Step 5: Run Post-Deploy SQL

```
mysql -h <RDS_HOST> -u moodleuser -p airpayprod < /tmp/academy-release/config/post-deploy.sql
```

This sets:

- Theme to airpayux
- Enables global search (simplifiedb)
- Disables perfdebug (removes debug text)
- Fixes catalog permissions for admin/manager roles

Step 6: Configure SMTP (if not already done)

Site Admin > Plugins > Message outputs > Email:

- SMTP hosts: your-smtp-server:587
- SMTP security: TLS
- SMTP username: academy@airpay.co.in
- SMTP password: (from IT team)

Step 7: Purge Caches and Disable Maintenance

```
# Purge all caches
sudo -u www-data php /var/www/html/admin/cli/purge_caches.php

# Clear theme cache specifically
sudo rm -rf /var/moodledata/localcache/theme/

# Disable maintenance mode
sudo -u www-data php /var/www/html/admin/cli/maintenance.php --disable

# Resume cron
# (restore crontab entry)
```

Step 8: Hard Refresh and Test

Tell all users to press Ctrl+Shift+R to clear browser cache.

4. Post-Install Configuration

Required Settings (Site Admin)

Setting	Path	Value
Theme	Appearance > Themes > Theme selector	airpayux
Default homepage	Front page > Front page settings	1 (Dashboard)
Enable web services	Plugins > Web services > Overview	Yes
Auth: login via email	Plugins > Authentication > Manage	Yes
Cron	Server > Scheduled tasks	Every 1 minute
Debug	Development > Debugging	NONE (for production)
Performance info	Development > Debugging > perfdebug	0

SMTP Configuration

Setting	Value
SMTP hosts	(from IT team)
SMTP security	TLS
SMTP auth type	LOGIN
SMTP username	academy@airpay.co.in
No-reply address	noreply@airpay.co.in

Cron Job Setup

```
# Add to crontab (run every minute)
* * * * * /usr/bin/php /var/www/html/admin/cli/cron.php > /dev/null 2>&1
```

5. Verification Checklist (21 Points)

After deployment, verify each item:

Authentication

- ☐ 1. Login page renders (split-screen layout with stats)
- ☐ 2. Admin can login (academy@airpay.co.in)
- ☐ 3. Employee can login
- ☐ 4. "Forgot Password" sends email (SMTP required)
- ☐ 5. Registration form works (5 fields)

Dashboards

- ☐ 6. Siteadmin sees: KPIs + Charts + Quick Nav + System Health + User Analytics
- ☐ 7. L&D Admin sees: KPIs (tenant-scoped) + Charts + Quick Nav (NO System Health)
- ☐ 8. Manager sees: Team KPIs + Compliance Table + Learner sections
- ☐ 9. Employee sees: Stats + Continue Learning + Timeline + Achievements + Recommendations
- ☐ 10. External user sees: Same as employee, tenant-isolated

Admin Pages

- ☐ 11. Manage Users loads with user cards
- ☐ 12. Manage Courses loads with course cards
- ☐ 13. Manage Company shows all 3 tenants (Airpay, Public, ZEEA)
- ☐ 14. Reports page loads
- ☐ 15. Online Exams page loads
- ☐

16. Classrooms page loads

Theme

- ☐ 17. Dark mode toggle works and persists across pages
- ☐ 18. Footer shows: logo, Privacy, Terms, Help, Contact, copyright
- ☐ 19. No debug text visible below footer
- ☐ 20. Mobile layout works at 590px viewport

Content

- ☐ 21. Course catalog loads with filters and course cards

6. Rollback Procedure

If anything breaks after deployment:

Quick Rollback (theme only)

```
# Re-enable old epsilon theme
mysql -h <RDS_HOST> -u moodleuser -p airpayprod -e "
    UPDATE mdl_config SET value = 'epsilon' WHERE name = 'theme';
"
sudo -u www-data php /var/www/html/admin/cli/purge_caches.php
```

Full Rollback (all files + database)

```
# 1. Enable maintenance mode
sudo -u www-data php /var/www/html/admin/cli/maintenance.php --enable

# 2. Restore files
sudo tar -xzf /backup/moodle-files-pre-upgrade-YYYYMMDD.tar.gz -C /

# 3. Restore database
mysql -h <RDS_HOST> -u moodleuser -p airpayprod < /backup/airpayprod-pre-upgrade-YYYYMMDD.sql

# 4. Purge caches
sudo -u www-data php /var/www/html/admin/cli/purge_caches.php
sudo rm -rf /var/moodledata/localcache/theme/

# 5. Disable maintenance mode
sudo -u www-data php /var/www/html/admin/cli/maintenance.php --disable
```

7. Troubleshooting

Common Issues

Issue: White screen after theme change

- Fix: `php admin/cli/purge_caches.php` + delete `/var/moodledata/localcache/theme/`
- If still broken: rollback theme to epsilon via SQL

Issue: "Table mdl_xxx doesn't exist"

- Fix: Run `php admin/cli/upgrade.php` to complete DB migration

Issue: Icons show as squares

- Fix: Hard refresh browser (Ctrl+Shift+R) + purge caches

Issue: jQuery "\$ is not a function" errors

- Fix: Verify BizLMS mustache fixes were applied (10 template files)

Issue: Catalog shows "No permissions"

- Fix: Run `post-deploy.sql` to fix viewcatalog capability for admin/manager roles

Issue: Manager doesn't see team view

- Fix: This is expected if manager has no Moodle capability roles. The dashboard detects managers via `open_supervisorid` field (direct reports). Verify the manager has direct reports in the user table.

Issue: "Reactive instances" text below footer

- Fix: Set `perfdebug` to 0 in `post-deploy.sql`

Issue: SCORM content doesn't play

- Fix: Ensure moodledata directory contains SCORM packages. These are NOT in the code package — they live in moodledata on the server.

Support Contacts

Role	Contact
L&D / Product Owner	Nitin Rajput (nitin.rajput@airpay.co.in)
IT Operations	(IT team lead email)
Database Admin	(DBA email)